National University of Computer and Emerging Sciences

0027

Lahore Campus

Electronic Circuit Design (EE2001)

Sessional-II Exam

Date: November 5th 2024

Course Instructor

Dr. Omer Saleem

Total Time (Hrs):	1
Total Marks:	30
Total Questions:	3

Roll No   Section   Student Signature

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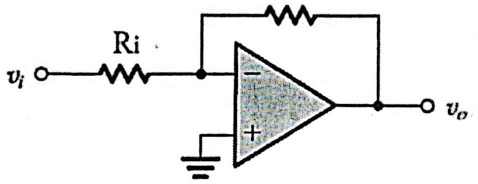
CLO # 3: Appraise the limitations of real op-amp circuits.

Q1: An imperfect op-amp is connected as shown below, with a R_i = 100 kΩ and R_f = 10.0 MΩ. The output voltage is found to be +9.35 V when a non-zero DC input voltage v_i is applied, and +7.10 V when the input is grounded (i.e. $v_i = 0$). Find the magnitude of v_i and the offset voltage v_{offset}. [10 marks]

-0.0228   0.07



CLO # 4: Design wave-form generation circuits.

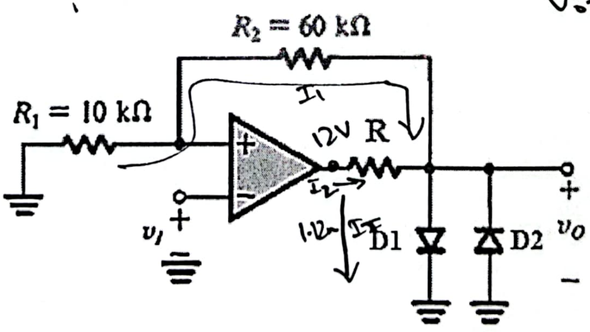
Q2: Refer to the following bistable multivibrator circuit. [10 marks]

a) Sketch and label the transfer characteristics of v_O–v_I. The diodes are assumed to have a constant 0.7 V drop when conducting, and the immediate output of the op-amp saturates at ±12 V.

b) Find the value of R, such that the maximum diode current in D1 is 1.12 mA when the op-amp output is saturated at +12 V?

1.716
-1.716

$V_o = 11.3$



$I_T = I_1 + I_2$

$1.12m = \frac{0 - V_o}{70K} + \frac{12 - V_o}{R}$

$1.12m + \frac{V_o}{70K} = \frac{12 - V_o}{R}$

$R = \frac{12 - V_o}{1.12m + \frac{V_o}{70K}}$

$= \frac{12 - 11.3}{1.12m + \frac{11.3}{70K}}$

$= 546.26\,\Omega$

CLO # 4: Design wave-form generation circuits.

Q3: Design a Wien-Bridge oscillator that generates a sine wave of frequency 9.95 kHz. [10 marks]

a) Draw the circuit schematic.

b) Find the values of all circuit components, using a minimum resistance of 150 kΩ in the circuit. R_f R_i?

$i = i_2 + i_1$